20260430 nGIST v7.2.1 and v7.6.8 comparison

The previous `nGIST` version we used is v7.2.1, and the new run is on v7.6.8. Here I summarize the major technical, physical, mathematical, and astrophysical differences between these versions, with emphasis on the expected changes in product maps.

The aim here is not to list cosmetic changes such as file renaming or keyword renaming. Instead, I focus on changes that may alter the actual numerical products: stellar kinematics, emission-line fluxes, gas-phase metallicities, BPT classification, Hα-based SFR, and stellar-population/SFH quantities.

Note on version number: The official release/website version used here is **v7.6.8**. A later GitHub tag labelled `v7.6.9` appears to exist, but for this comparison I treat **v7.6.8** as the relevant new version, because that is the latest official release referred to by the nGIST release page/website and is the version used for the new run.

1. Executive summary

The transition from `nGIST` v7.2.1 to v7.6.8 is not just a cosmetic update. Several core fitting choices changed, especially in the `KIN` and `SFH` modules. The `GAS` pPXF emission-line fitting model is broadly similar, but its products can still change indirectly because of updates to the input spectra, noise treatment, stellar-continuum treatment, and fixed stellar kinematics.

The most important result-level expectation is:

> *Strong emission-line fluxes such as Hα and [N II] are expected to remain broadly stable in high-S/N regions, but Hβ, [O III]/Hβ, O3N2 metallicity, BPT masks, stellar velocity dispersion, h3/h4, SFH weights, and low-S/N/spaxel-level products can change.*

The highest-impact changes are:

MODULE	MAIN TECHNICAL CHANGE	EXPECTED RESULT IMPACT
`KIN`	Updated pPXF dust treatment, optimal-template selection, noise handling, and outlier clipping	Stellar kinematics can change, especially σ, h3/h4, and continuum-sensitive regions
`GAS`	Core pPXF emission-line call remains similar, but SPAXEL/BOTH-mode bugs and input/output handling were fixed	Strong-line fluxes likely stable in clean regions; SPAXEL/BOTH products and weak lines may change
`SFH`	Updated pPXF dust treatment, template selection, clipping, and MC treatment	Stellar-population maps can change substantially
`READ_DATA` / pipeline input	Later versions include corrected noise handling and Galactic extinction correction in MUSE read modules	Formal errors, χ^2, weak-line fitting, and continuum fits may change
BPT maps	Built-in BPT-map construction changed	Recompute BPT classification yourself from raw fluxes for science use

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2. KIN module: expected impact on stellar kinematics and continuum modelling

The `KIN` module is one of the most important sources of version-dependent changes.

In v7.2.1, the KIN workflow was closer to:

1. Fit a combined/mean spectrum to obtain an optimal stellar template.
2. Use that optimal template in intermediate per-bin fits.
3. Estimate the noise from residuals.
4. If `DOCLEAN=True`, use pPXF's internal `clean=True` option.
5. Run a final pPXF fit using the full template library.
6. Save stellar kinematics and best-fit spectra.

In v7.6.8, the workflow is more complex:

1. A first pPXF fit is used to obtain an initial optimal template or template set.
2. A step-0 pPXF dust fit is performed, generally without additive or multiplicative polynomials.
3. The fitted dust can then be fixed in later pPXF steps.

4. The optimal-template logic can use a set of non-zero-weight templates rather than a single combined optimal template.
5. The outlier-rejection step no longer relies simply on pPXF `clean=True`; instead, a custom `clip_outliers()` mask is used.
6. The final fit uses the updated noise, updated mask, and optionally fixed dust.

The physical meaning of this change is that v7.6.8 has a more controlled separation between:

- stellar continuum shape,
- dust attenuation / spectral slope,
- multiplicative-polynomial correction,
- template mismatch,
- masked pixels,
- residual-based noise rescaling.

This is scientifically relevant because the stellar kinematics are not measured from isolated lines. They are inferred from a full-spectrum fit, so the continuum shape, masked spectral regions, template mixture, and weighting scheme can shift the best-fit velocity dispersion and higher-order moments.

Expected changes in KIN products

PRODUCT	EXPECTED CHANGE	REASON
`V`	Usually small	Stellar velocity is relatively robust if the old fit converged well
`SIGMA`	Small to moderate	More sensitive to template mismatch, masking, and continuum shape
`H3`, `H4`	Potentially noticeable	Higher moments are sensitive to residual structure and bad-pixel rejection
`SNR_POSTFIT`	Likely changed	Noise estimation and residual treatment changed
`RED_CHI2` / χ^2	Likely changed	The effective noise and mask can change
Best-fit continuum	Can change	New dust/template/masking treatment
Spectral mask / clipped pixels	Can change	pPXF `clean=True` replaced by custom clipping

KIN matters mainly because the emission-line module may use the stellar kinematics as fixed input. If `GAS.FIXED=True`, then a change in stellar kinematics can propagate into the gas fit. This is especially relevant near Hβ, where stellar Balmer absorption must be modelled accurately.

The most vulnerable quantities are:

- Hβ flux,
- Hβ equivalent width,
- [O III]/Hβ.

The least vulnerable quantities are:

- Hα flux,
- [N II]/Hα,
- [S II]/Hα.

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3. GAS module: core pPXF fitting is similar, but some products can still change

The `GAS` module requires a more careful interpretation.

The basic pPXF emission-line fitting call is broadly similar between v7.2.1 and v7.6.8. Both versions fit stellar templates and gas templates together, use pPXF gas templates, use tied/fixed gas components, and return pPXF gas fluxes and errors.

Therefore, if the following are identical:

- input spectra,
- error spectra,
- wavelength range,
- spectral mask,
- LSF,
- stellar templates,
- gas templates,
- fixed stellar kinematics,
- multiplicative polynomial degree,
- emission-line configuration,

then the strong-line gas fluxes should be broadly comparable.

However, in practice these conditions are not guaranteed. The GAS result can change because the upstream products and module behaviour changed.

3.1 Strong-line fluxes

For high-S/N regions, I expect the following to be relatively stable:

LINE	EXPECTED STABILITY
Hα	high
[N II] λ6583	high
[S II] λ6716,6731	moderate to high
[O III] λ5007	moderate to high, depending on S/N
Hβ	moderate; more sensitive to continuum modelling

Hα and [N II] are close in wavelength and usually strong, so their ratio should be robust. Hβ is weaker and sits on stellar Balmer absorption, so it is more sensitive to changes in continuum modelling.

3.2 SPAXEL and BOTH mode

The version history indicates that several GAS bugs were fixed after v7.2.1, especially for `SPAXEL` and `BOTH` modes. This is important.

If the old v7.2.1 products were produced only at `BIN` level, the comparison is more likely to be stable.

If the old products were produced at `SPAXEL` or `BOTH` level, I would not assume they are reliable without checking. v7.6.8 should be preferred for new spaxel-level products.

3.3 Noise and variance handling

Later versions mention correction of "noise instead of variance" in the read-data step. This matters because pPXF uses the input noise vector as a weight.

Mathematically, a global multiplicative rescaling of all noise values does not change the χ^2 minimum. But a wavelength-dependent change in the noise vector can change the best-fit solution, because different spectral regions receive different weights.

This can affect:

- formal line-flux errors,
- χ^2,
- weak-line fluxes,
- continuum shape around weak features,
- low-S/N outskirts,
- sky-contaminated wavelength regions.

The strongest impact is expected for weak or continuum-sensitive lines:

- Hβ,
- [O III] λ5007 in weak-excitation regions,
- [O I] λ6300 if used,
- [S III] λ6312 if used,
- auroral/direct-Te lines if attempted.

3.4 Built-in BPT maps

The built-in BPT map products should be treated cautiously.

For the paper analysis, it is safer to recompute BPT classification directly from the raw line-flux columns, using our own cuts and error requirements. This is especially important because our science analysis uses a conservative H II-region selection, requiring robust line detections and error bars to stay within the H II region on both [N II] and [S II] BPT diagrams.

Recommended practice:

```
Do not rely blindly on the built-in NII_Ha_BPT or SII_Ha_BPT
columns.
Use the flux columns directly and recompute BPT classification
externally.
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This also keeps the classification method independent of any internal implementation changes in the pipeline.

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4. SFH module: expected to change more strongly than GAS

The `SFH` module is the most version-sensitive of the main science modules.

The output SFH quantities are not direct observables. They are inferred from template weights in a high-dimensional stellar population fit. These weights are sensitive to many details:

- dust treatment,
- multiplicative polynomial degree,
- emission-line masking,
- whether emission-cleaned spectra are used,
- fixed or free stellar kinematics,
- noise vector,
- outlier clipping,
- regularisation,
- template library,
- wavelength range.

Between v7.2.1 and v7.6.8, several of these changed or became more flexible.

Expected changes in SFH products

PRODUCT	EXPECTED CHANGE	REASON
Mean stellar age	Can change substantially	Age-dust-metallicity degeneracy and updated dust handling
Stellar metallicity	Can change	Sensitive to template weights and continuum treatment
α-enhancement	Can change	Sensitive to template library and line-index-like residuals
Template weights	Can change substantially	Full-spectrum template-weight solution is non-unique
MC uncertainties	Can change	The MC procedure uses the updated noise/mask/model
SFH χ^2	Likely changed	Noise and mask changed
SFH best-fit continuum	Can change	Updated dust and clipping

For the current MAUVE-MUSE gas-phase metallicity/SFR work, the SFH module only matters if we use nGIST-derived stellar population products or nGIST-derived stellar mass surface-density maps.

If the stellar mass surface density, SFR, and metallicity products are instead taken from the existing MAUVE data products or from separate calibrated maps, then SFH changes are less central to the current paper.

However, for future work involving stellar populations, age gradients, or stellar metallicity, v7.2.1 and v7.6.8 SFH maps should not be mixed.

Source links for follow-up

- nGIST releases: https://github.com/geckos-survey/ngist/releases
- nGIST v7.6.8 release: https://github.com/geckos-survey/ngist/releases/tag/V7.6.8
- nGIST changelog: https://github.com/geckos-survey/ngist/blob/V7.6.8/CHANGELOG
- GitHub comparison: https://github.com/geckos-survey/ngist/compare/V7.2.1...V7.6.8
- pPXF documentation / package page: https://pypi.org/project/ppxf/